5.5 Solving Rational Inequalities

A Rational Inequalities

In order to solve a rational (nonlinear) inequality:

1. state *restrictions*
2. *move* all the terms to one side
3. find the *LCD* (Least Common Denominator) and simplify the rational expression
4. *factor* both the numerator and the denominator
5. find the *sign* of the rational expression by using a *sign chart*, *graph* or *critical numbers* method
6. *conclude* and verify if *restrictions* are satisfied

Ex 1. Is possible to use cross multiplication to solve the following inequality? Explain. Solve it by using four different methods.

$\frac{1}{x} \le \frac{2}{x+1}$ No !!

CANNOT MULTIPLY VARIABLES OVER THE INEQUALITY SIGN

Ex 2. Solve the following inequalities:

a) $\frac{x+1}{x-1} \ge 0$

SIGN CHART

		-1		1	
$x+1$	$-$		$+$		$+$
$x-1$	$-$		$-$		$+$
$\frac{x+1}{x-1}$	$+$	0	$-$	$\nexists$	$+$

$\therefore x \in (-\infty,-1] \cup (1,\infty)$

b) $\frac{x^2-1}{x-2} \le 0$

$\frac{(x-1)(x+1)}{(x-2)} \le 0$

SIGN CHART

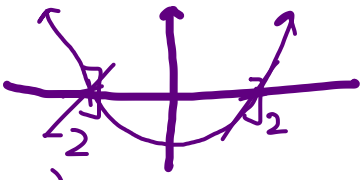
		-1		1		2	
$x+1$	$-$		$+$		$+$		$+$
$x-1$	$-$		$-$		$+$		$+$
$x-2$	$-$		$-$		$-$		$+$
$f(x)$	$-$	0	$+$	0	$-$	$\nexists$	$+$

$\therefore x \in (-\infty,-1] \cup [1,2)$

c) $\frac{x^2+1}{x^2-4} > 0$

PSEUDO "GRAPH"

$\frac{x^2+1}{(x-2)(x+2)} > 0$

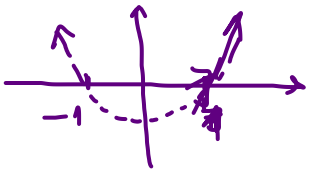


$x \in (-\infty,-2) \cup (2,\infty)$

d) $\frac{x^3+1}{x^3-1} < 0$

PSEUDO GRAPH

$\frac{(x+1)(x^2-x+1)}{(x-1)(x^2+x+1)} < 0$



$\therefore x \in (-1,1)$

OR CRITICAL NUMBERS

$\frac{(x+1)(x^2-x+1)}{(x-1)(x^2+x+1)}$

Ex 3. Solve the following inequalities:

a) $\frac{1}{x-1} > \frac{1}{x+1}$

$\frac{1}{x-1} - \frac{1}{x+1} > 0$ $\quad x \ne 1, -1$

$\frac{x+1-x+1}{(x-1)(x+1)} > 0$

$\frac{2}{(x-1)(x+1)} > 0$

$\therefore \boxed{x \in (-\infty,-1) \cup (1,\infty)}$

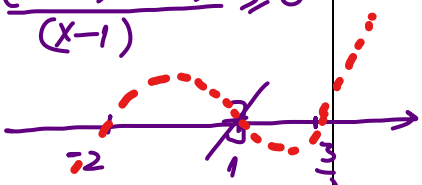
b) $4x - \frac{5}{x-1} \ge 2x - 1$

$4x - \frac{5}{x-1} - 2x + 1 \ge 0$

$\frac{(2x+1)(x-1)-5}{(x-1)} \ge 0$

$\frac{2x^2+x-6}{x-1} \ge 0$

$\frac{(x+2)(2x-3)}{(x-1)} \ge 0$ $\quad x \ne 1$



$\therefore \boxed{x \in [-2,1) \cup [\tfrac{3}{2},\infty)}$

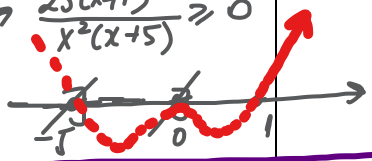
c) $\frac{4x+5}{x^2} \ge \frac{4}{x+5}$

$x \ne 0, -5$

$\frac{4x+5}{x^2} - \frac{4}{x+5} \ge 0$

$\frac{4x^2+25x+25-4x^2}{x^2(x+5)} \ge 0$

$\frac{25(x+1)}{x^2(x+5)} \ge 0$



$\therefore \boxed{x \in (-\infty,-5) \cup [1,\infty)}$

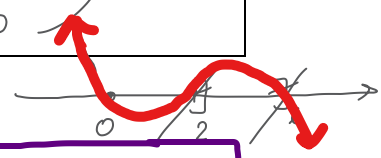
d) $\frac{x}{2x-4} - \frac{3}{x-6} \le \frac{1}{2}$

$\frac{x}{2(x-2)} - \frac{3}{x-6} - \frac{1}{2} \le 0 \; ; x \ne 2, 6$

$\frac{x(x-6) - 6(x-2) - (x-2)(x-6)}{2(x-2)(x-6)} \le 0$

$\frac{\cancel{x^2} - 6x - 6x + 12 - \cancel{x^2} + 8x - 12}{2(x-2)(x-6)} \le 0$

$-\frac{2x}{(x-2)(x-6)} \le 0$



$\therefore \boxed{x \in [0,2) \cup (6,\infty)}$

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Ex 4. Solve the following inequality:

$$\frac{x}{x-2}+\frac{1}{x-4}\geq\frac{2}{x^2-6x+8}$$

$x\neq 2,4$

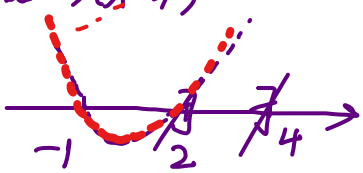
$$\frac{x}{x-2}+\frac{1}{x-4}-\frac{2}{(x-2)(x-4)}\geq 0$$

$$\frac{x(x-4)+x-2-2}{(x-2)(x-4)}\geq 0$$

$$\frac{x^2-4x+x-4}{(x-2)(x-4)}\geq 0$$

$$\frac{x(x-4)+(x-4)}{(x-2)(x-4)}\geq 0$$

$$\frac{(x+1)\cancel{(x-4)}}{(x-2)\cancel{(x-4)}}\geq 0,\ x\neq 2,4$$



$\therefore\ x\in(-\infty,-1]\cup(2,4)\cup(4,\infty)$

Ex 5. Solve the following inequality:

$x\neq 0,1,2$

$$\frac{5}{x}\leq\frac{6}{x-1}<\frac{x}{x-2}$$

DOUBLE INEQUALITY

$\frac{5}{x}\leq\frac{6}{x-1}$ and $\frac{6}{x-1}<\frac{x}{x-2}$

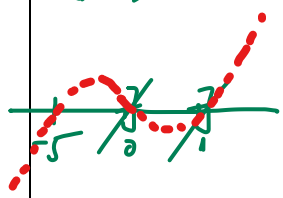
$$\frac{5}{x}-\frac{6}{x-1}\leq 0$$

$$\frac{5x-5-6x}{x(x-1)}\leq 0$$

$$\frac{-x-5}{x(x-1)}\leq 0$$

OR

$$\frac{x+5}{x(x-1)}\geq 0$$



$\therefore\ x\in[-5,0)\cup(1,\infty)$

$$\frac{6}{x-1}-\frac{x}{x-2}<0$$

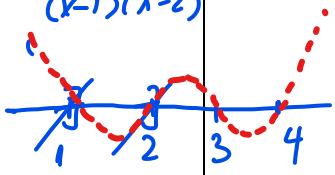
$$\frac{6x-12-x^2+x}{(x-1)(x-2)}<0$$

$$\frac{-x^2+7x-12}{(x-1)(x-2)}<0$$

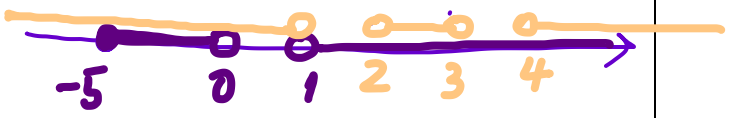
OR

$$\frac{x^2-7x+12}{(x-1)(x-2)}>0$$

$$\frac{(x-4)(x-3)}{(x-1)(x-2)}>0$$



and $\therefore\ x\in(-\infty,1)\cup(2,3)\cup(4,\infty)$



$\therefore\ x\in[-5,0)\cup(2,3)\cup(4,\infty)$

Ex 6. Solve the following inequality:

$$\frac{1}{|x-1|}-\frac{|x+1|}{x}\leq 2$$

OPTIONAL

Reading: Nelson Textbook, Pages 288-295
Homework: Nelson Textbook, Page 295: #4ab, 5acf, 7, 12, 13

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